Audio lecture two. Amniotic fluid physiology, ultrasound assessment, oligohydramnios, and polyhydramnios.

Companion reading handout for simultaneous listening.
This lecture is based mainly on Gabbe, chapter 35, Amniotic Fluid Disorders, Gabbe, chapter 9, Obstetric Ultrasound, and Israeli Position Paper number 37 on polyhydramnios and oligohydramnios.

The most important idea

Let us begin with the most important idea.

Amniotic fluid is not passive fluid around the fetus. It is a dynamic physiologic compartment. It is produced, swallowed, exchanged, measured, mismeasured, and interpreted. When it is abnormal, it can be a sign of fetal anatomy, placental function, membrane integrity, maternal disease, or fetal swallowing.

If you remember only one image, picture a lake with several rivers flowing into it and several drains flowing out. The level of the lake is the amniotic fluid volume. The rivers are fetal urine and lung liquid. The drains are fetal swallowing and intramembranous absorption. Disease changes the inflow, the outflow, or the container.

Building the system

Let us build that system from the beginning.

Amniotic fluid volume rises through pregnancy and peaks around the early third trimester. Gabbe describes a wide normal range, especially around **32 to 33 weeks**. This wide range is important. It means amniotic fluid is regulated, but not with the precision of serum sodium. There is real biologic variation and real measurement variation.

Fetal urine

In the second half of pregnancy, the dominant source of amniotic fluid is fetal urine. Human fetal kidneys begin producing urine before the end of the first trimester, and urine production increases with gestational age. Near term, human fetal urine production is roughly **1000 to 1200 milliliters per day**. That is an astonishing number. It means the amniotic fluid compartment is not stagnant. It is being renewed continuously.

This is why renal agenesis, severe obstructive uropathy, or severe uteroplacental insufficiency can cause oligohydramnios. If the kidneys do not exist, cannot drain, or are underperfused, the main river into the lake dries up.

Fetal lung liquid

The second contributor is fetal lung liquid. The fetal lung is not just waiting for birth. It actively produces fluid that exits the trachea. Some is swallowed, and some enters the amniotic cavity. The presence of surfactant in amniotic fluid near term is one clue that lung liquid moves outward. This outward flow also helps maintain lung expansion and development.

Removal of amniotic fluid

Now we transition to removal.

Fetal swallowing

Fetal swallowing begins early and increases with gestational age. The fetus swallows amniotic fluid, and the fluid then moves through the gastrointestinal tract. If the fetus cannot swallow effectively, fluid accumulates. This is why central nervous system disorders, neuromuscular disease, and gastrointestinal obstruction can produce polyhydramnios.

Duodenal atresia is a classic example. Gabbe chapter 9 describes the double bubble sign, and polyhydramnios is common because fluid reaches the fetal mouth, but cannot pass normally through the proximal gastrointestinal tract.

Intramembranous absorption

But swallowing is not enough to explain normal fluid balance. This is where the most elegant concept enters: **intramembranous absorption**.

Intramembranous absorption is the movement of water and solutes from the amniotic compartment into fetal blood circulating on the fetal surface of the placenta. Gabbe chapter 35 frames it as the missing link. If fetal urine and lung liquid entered the amniotic space and swallowing were the only drain, there would be hundreds of milliliters per day of excess fluid. Yet acute polyhydramnios does not occur in normal pregnancy. Therefore another drain must exist.

That drain is the intramembranous pathway.

Under normal conditions, about **200 to 500 milliliters per day** may leave through this route. Under experimental conditions in sheep, intramembranous absorption can increase almost tenfold. The concept is memorable because it turns the placenta into part of the amniotic fluid regulator. The amnion is not just a bag, and the placenta is not just a gas exchanger. The fetal surface of the placenta participates in fluid traffic.

Physiology summary

Let us pause and summarize physiology.

- Fetal urine is the main inflow in the second half of pregnancy.
- Lung liquid also contributes.
- Fetal swallowing removes fluid.
- Intramembranous absorption is the hidden drain through the fetal placental surface.
- Oligohydramnios usually means too little inflow, membrane rupture, or placental insufficiency.
- Polyhydramnios usually means too much production, impaired swallowing, fetal anomaly, diabetes, anemia, hydrops, or twin-related pathology.

Pause here if you want to review the lake model before moving to ultrasound measurement.

Ultrasound measurement

Now, how do we measure amniotic fluid?

True volume can be measured by dye dilution at amniocentesis, but that is not practical for routine care. So ultrasound is used.

There are two common semiquantitative methods.

Amniotic Fluid Index, or A F I

The first is the Amniotic Fluid Index, or A F I. The uterus is divided into four quadrants. In each quadrant, the deepest vertical pocket is measured, avoiding fetal parts and cord. The four measurements are added. A common definition is oligohydramnios when A F I is less than **5 centimeters**, and polyhydramnios when A F I is greater than **24 or 25 centimeters**.

Maximum vertical pocket, deepest vertical pocket, M V P, or D V P

The second method is the maximum vertical pocket, also called the deepest vertical pocket, or M V P, or D V P. This is the single deepest pocket free of fetal parts and cord. In many guidelines, oligohydramnios is diagnosed when the D V P is less than **2 centimeters**. Polyhydramnios is diagnosed when the deepest pocket is greater than **8 centimeters**.

Israeli Position Paper number 37 describes both methods. It emphasizes that subjective assessment is part of every ultrasound, while objective measurement should be performed when the subjective impression is abnormal. It also notes that in dichorionic twins, M V P is the simplest practical method for each sac.

A F I versus D V P

Now, a key exam concept: **A F I versus D V P.**

A F I tends to diagnose more oligohydramnios. That sounds good until you ask the outcome question. Studies summarized in Gabbe show that A F I increases the diagnosis of oligohydramnios, increases induction and intervention, and does not improve perinatal outcomes compared with D V P. Therefore, D V P is often preferred for clinical decisions, especially to avoid unnecessary delivery.

The memory hook is: **A F I is more sensitive to anxiety. D V P is often more clinically disciplined.**

Oligohydramnios

Now let us turn to oligohydramnios.

Oligohydramnios is not one disease. It is a sign. The differential diagnosis changes dramatically by gestational age.

Before viability

Before viability, severe oligohydramnios is ominous because the fetal lungs need fluid and space to develop. The canalicular phase of lung development is especially vulnerable. Without adequate fluid, the chest is compressed, lung liquid dynamics are altered, terminal bronchioles and alveolar development suffer, and lethal pulmonary hypoplasia can result.

Gabbe chapter 9 describes the oligohydramnios sequence: pulmonary hypoplasia, fetal deformations, and flexion contractures. Position Paper number 37 similarly emphasizes that second trimester oligohydramnios, especially before **24 weeks**, is associated with preterm birth, pulmonary hypoplasia, and contractures.

The cause matters

The cause matters.

If severe oligohydramnios is due to bilateral renal agenesis, the prognosis is essentially fatal because the fetus cannot produce urine and the lungs cannot develop normally. Gabbe chapter 35 states that with renal agenesis, virtually all newborns die because of pulmonary hypoplasia. This is why a midtrimester scan with absent kidneys, absent bladder, and anhydramnios is not just a renal diagnosis. It is a lung development diagnosis.

If the cause is premature rupture of membranes after viability, prognosis depends on gestational age, infection, latency, and residual fluid. The earlier the rupture and the longer the severe oligohydramnios, the higher the risk of pulmonary hypoplasia and deformities.

If the cause is placental insufficiency, the fetus may be growth restricted. Chronic hypoxemia causes redistribution away from the fetal kidneys. Less renal perfusion means less urine. Less urine means less fluid. Here the low fluid is a marker of chronic adaptation to placental disease.

If the cause is maternal dehydration or hypovolemia, fluid can sometimes improve with maternal hydration, but this should not distract from evaluating fetal anatomy, growth, membrane status, and placental disease.

Logical workup

So when oligohydramnios is diagnosed, the workup should be logical.

1. First, ask whether the membranes are ruptured. The history may include leakage, wetness, or clear fluid. A sterile speculum exam can look for pooling, ferning, and alkaline pH, and commercial protein tests may help when the diagnosis is unclear.
2. Second, evaluate fetal anatomy, especially kidneys and bladder. If the kidneys and bladder are present and the fetus is normally grown, ruptured membranes becomes more likely.
3. Third, evaluate growth and placenta. Growth restriction, hypertension, antiphospholipid syndrome, or abnormal placental appearance shifts the concern toward uteroplacental insufficiency.
4. Fourth, interpret gestational age. The same D V P at 38 weeks and at 18 weeks does not carry the same implications.

Oligohydramnios summary

Let us pause and summarize oligohydramnios.

Low fluid is a symptom, not a diagnosis. Think rupture, renal or urinary tract anomaly, placental insufficiency, prolonged pregnancy, growth restriction, and maternal conditions. Early severe oligohydramnios threatens lung development. Late isolated oligohydramnios can be overdiagnosed, especially by A F I, and should be interpreted carefully.

Polyhydramnios

Now let us transition to polyhydramnios.

Polyhydramnios is too much fluid. It may be mild and idiopathic, or it may be the clue to a major fetal or maternal problem.

Gabbe chapter 9 classifies polyhydramnios as mild when A F I is above **24 centimeters** or deepest pocket is above **8 centimeters**. Moderate is higher, and severe is higher still, with severe polyhydramnios raising a much stronger concern for fetal anomaly.

The physiology is the mirror image of oligohydramnios. Too much fluid can result from increased fetal urination, impaired swallowing, abnormal absorption, or conditions that increase fetal cardiac output or hydrops.

Maternal diabetes

Maternal diabetes is a classic cause. Fetal hyperglycemia causes osmotic diuresis. The fetus urinates more, so the lake rises. Polyhydramnios plus macrosomia should always make you think of diabetes or poor glycemic control.

Fetal gastrointestinal obstruction

Fetal gastrointestinal obstruction is another classic cause. If the fetus cannot swallow or absorb amniotic fluid normally, the outflow drain is blocked. Duodenal atresia, esophageal atresia, neurologic impairment, and neuromuscular disease all fit this logic.

Fetal anemia, hydrops, and twin-related pathology

Fetal anemia and hydrops can also cause polyhydramnios, through high-output physiology and altered fluid handling. Twin-to-twin transfusion syndrome is another important cause in monochorionic twins, although Position Paper number 37 explicitly excludes monochorionic twin management from its scope.

Position Paper number 37 notes that most polyhydramnios is idiopathic, but the more severe it is, the more likely a significant abnormality will be found. Severe polyhydramnios is not just more water. It is a stronger signal.

Management

Management of polyhydramnios therefore begins with cause. Evaluate anatomy. Consider genetic conditions when anomalies are present. Screen or reassess for diabetes. Think about infection, anemia, hydrops, and multiple gestation. Follow growth and fluid. But isolated polyhydramnios itself is not automatically an indication for cesarean delivery, and it is not automatically an indication for early delivery before 39 weeks. Severe symptomatic cases may require individualized delivery timing.

Final synthesis

Now final synthesis.

1. **First, remember the lake.** Fetal urine and lung liquid fill it. Swallowing and intramembranous absorption drain it.

2. **Second, remember that fetal urine is the dominant source in the second half of pregnancy, reaching about 1000 to 1200 milliliters per day at term.**
3. **Third, remember intramembranous absorption.** It is the hidden placental drain, and it can increase almost tenfold experimentally.
4. **Fourth, remember measurement discipline.** A F I less than 5 centimeters or D V P less than 2 centimeters defines oligohydramnios, but D V P often prevents unnecessary intervention.
5. **Finally, remember gestational age.** Early severe oligohydramnios is a lung-development emergency. Late isolated low fluid is a surveillance and context problem, not an automatic panic button.

End of lecture two. Pause here before continuing to P PROM.